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Harnessing machine learning and synthetic microbial communities to control harmful algal blooms

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ABSTRACT

Freshwater with high quality is crucial for both public health and aquatic biodiversity. However, freshwater resources face numerous challenges, including the proliferation of harmful algal blooms (HABs) caused by various cyanobacterial species that are generally triggered by human activities like agricultural runoff and wastewater. Native algicidal microbiomes may offer potential solutions, although challenges remain in utilizing microbial resources to mitigate HABs in freshwater environments. The combination of synthetic microbial community and probiotic development approaches with robust machine learning tools could allow us to harness native microbiomes to address water quality issues caused by HABs in large water bodies. A meta-analysis of around 100 research studies regarding algicidal bacteria-algae interactions was conducted to quantitatively assess the potential of taxonomically diverse microbial species in controlling HABs in freshwater ecosystems. Meta-analysis findings revealed that diverse species from common freshwater bacterial phyla such as *Actinobacteria, Bacteroidota, Firmicutes*, and *Proteobacteria* exhibited 50100 % algicidal activity against different algal species depending on interacting species and environmental conditions. Algicidal taxa (mainly against *Microcystis aeruginosa*) from both *Actinobacteria* and *Firmicutes* primarily included Actinomycetes and Bacillus species. However, *Bacteroidota* and *alpha/beta Proteobacteria* exhibited algicidal activity against a broader range of algal species, thus highlighting their potential for controlling multi-species HABs in freshwater environments. Based on this quantitative analysis, the current review puts forward synthetic microbial communities and machine-learning based frameworks to develop microbial solutions for protecting freshwater resources from HABs invasions.

1. Introduction

High-quality freshwater is an essential resource for ensuring an adequate supply of water to meet domestic, industrial, and agricultural demands (Wetzel, 2000). Only 3 percent of the available freshwater is utilized by the global population where only 0.06 % of the freshwater is readily obtainable, with lakes comprising 87 % and rivers account for a mere of 2 % freshwater (El-Dessouky and Ettouney, 2002; Musie and Gonfa, 2023).Needless to mention that human society depends on freshwater resources as their primary water sources for multiple needs. However, human activities in last few decades including urbanization and industrialization, as well as other destructive initiatives such as developing reservoirs and inter-basin transfers, have been altering freshwater systems on a large scale (Carpenter et al., 2011; Malmqvist and Rundle, 2002; Meybeck, 2003). In addition climate change, particularly rising water temperatures, and land use alterations such as agriculture and deforestation, are directly and indirectly jeopardizing the health and functionality of freshwater ecosystems through increased agricultural and sediment runoff (Hemdan et al., 2024; Lake et al., 2000; Sala et al., 2000).

Among aquatic organisms, microbial communities or microbiomes perform critical roles in regulating the biological and chemical

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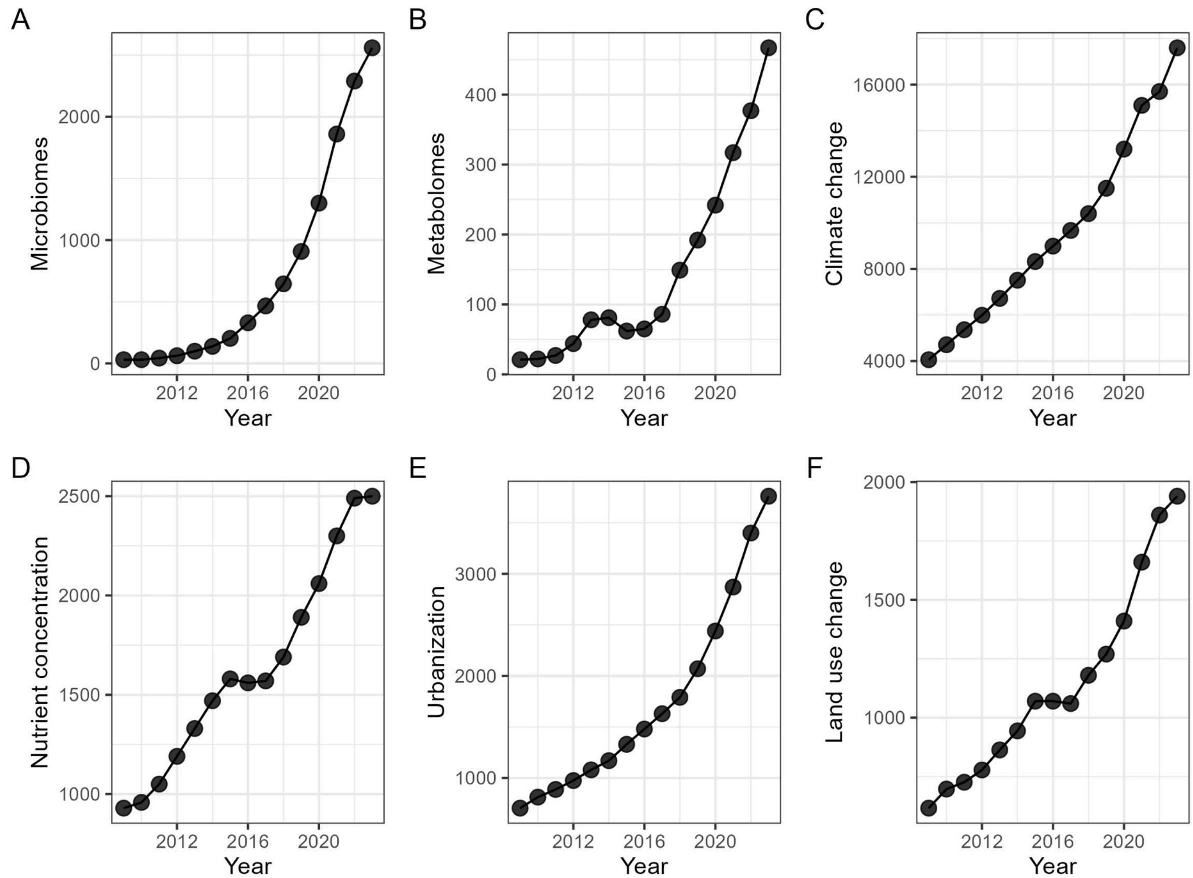

Fig. 1. Annual number of published scientific papers on HABs in relation to other biotic and abiotic factors (2009 to 2023).

properties of the water ecosystems. Microbes improve and sustain water quality by consuming, immobilizing, absorbing, and recycling essential nutrients such as carbon (C), nitrogen (N), phosphorus (P), sulfur (S), and toxic metals (Ashraf et al., 2023; Shmeis, 2018; Yousuf et al., 2022). Various microbial functional groups such as algae, bacteria, protists, and phages are known to regulate nutrient turnovers and water chemistry both at local and global scales (Kuhlisch et al., 2024; Newton et al., 2011; Sarker et al., 2019). During this period, the microbial communities of freshwaters are sensitive to several environmental conditions such as water pH, temperature, nutrient concentrations, organic matter and labile carbon contents, salinity, heavy metal contents, light intensity and duration, land use, and climatic changes (Weller et al., 2022; Zeglin, 2015). Both natural and anthropogenic disturbances of freshwater environments could also lead to several other ecological concerns related to environmental and human health. For instance, changes in water chemistry and contamination by human activities may lead to the emergence and development of unfavorable microbial taxa, for example, the blooms of harmful algae, thus compromising water quality for aquatic and human life (Rashidi et al., 2021; Wang et al., 2011).

The harmful algal blooms (HABs) mediated alterations in the freshwater environments like oxygen depletion, toxins from algae, and changes in water pH could potentially directly and indirectly assert significant effects on microbial diversity, from individual taxa to community level (Hale et al., 2015; Krausfeldt et al., 2024). Thus, it is very important to investigate the relationship of microbial taxa with the harmful algae, which are common in the freshwaters, and pose significant ecological and economic threats to the contemporary society (Abate et al., 2024; Igwaran et al., 2024). This is also important because only culturable abundance of bacteria ranges from 10^5 to 10^9 per Liter of freshwaters whereas the total abundance could be more than 100 folds than the former ones (Pound et al., 2021).

If the key regulators of harmful algae are put together, intriguingly there is a growing interest in studying the responses of freshwater environments with respect to their (i) microbiomes, (ii) metabolome (comprehensive collection of all low-molecular weight chemicals generated by the metabolic processes of a living organism (Heux et al., 2015)), (iii) climate change, (iv) nutrient concentration, and (v) land use change (Fig. 1). Researchers have demonstrated interest in various biotic factors such as microbiomes and metabolomes, as well as abiotic factors including climate change, elevated nutrient concentrations like nitrogen and phosphorus that contribute to eutrophication and land use changes that are responsible for algal blooms, publishing their findings in this domain.

In this regard, previous studies mostly depended on single-source data for predicting and managing HABs. For instance, most previous studies considered physical, biological, and climatic variables including day of the year, salinity, temperature, and upwelling index; another included elevated CO_2 and higher temperature as the factor for HABs development (Brandenburg et al., 2019; Vilas et al., 2014). These previous studies either relied on in-person or remote sensing data for prediction of HABs and thus don't provide information about complex environmental interactions underlying HABs in freshwater environments. The machine learning approaches, however, can efficiently manage complicated interactions among microbial and physicochemical variables, record temporal dynamics, process vast amounts of data, and handle non-linear correlations between predictors. These approaches can aggregate all real-time water quality data, climate data, land use data, historical HAB episodes, meteorological data, hydrological parameters and microbiomics profiles. As a result, these approaches could effectively help in predicting and mitigating HAB threats to the freshwater environments (Bergen et al., 2019). This quantitative review article aims to advance pre-existing and scattered understanding of

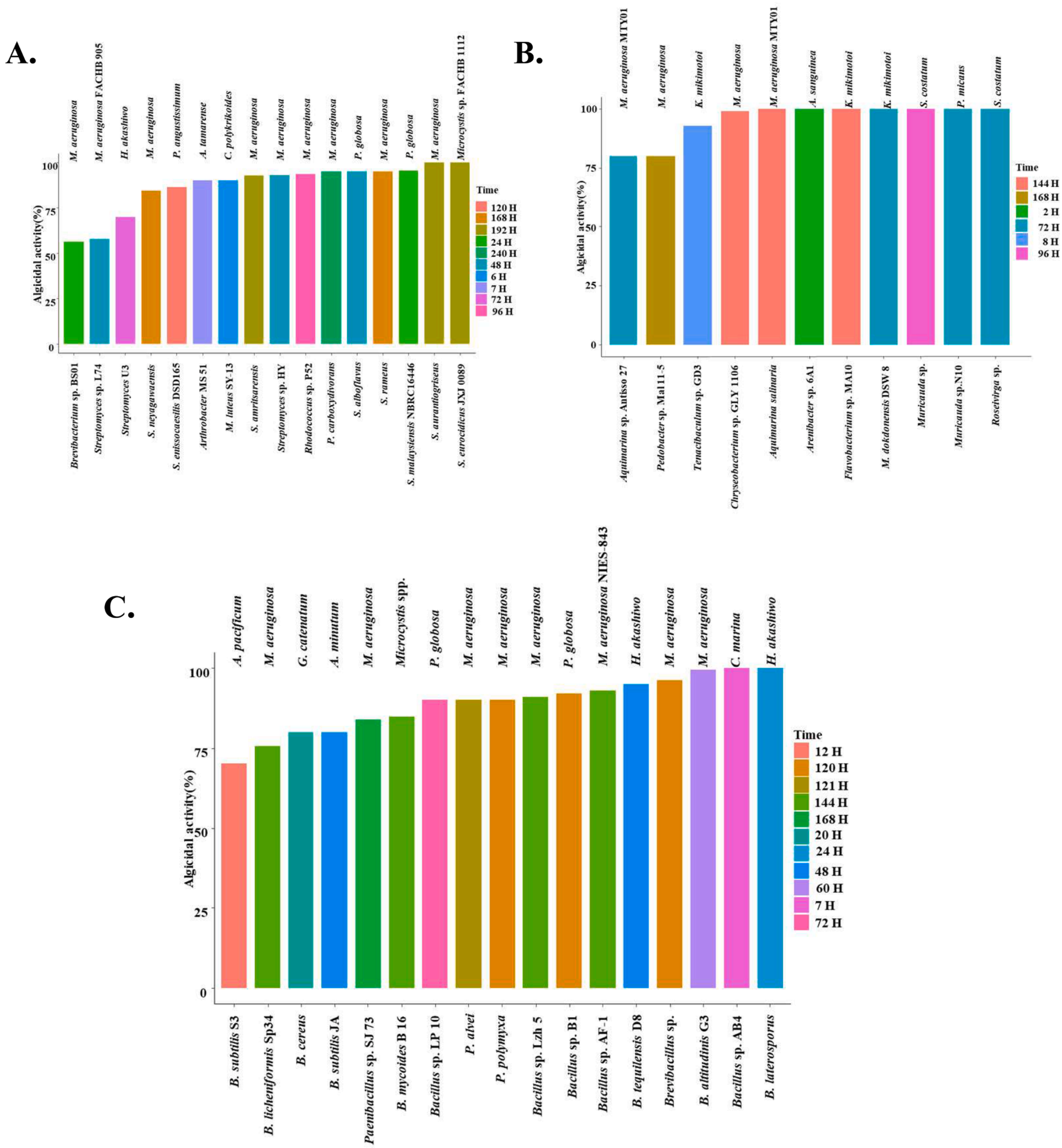

Fig. 2. Algicidal activity and efficacy of bacterial taxa belonging to actinobacteria (a), bacteriodetes (b) and firmicutes (c) against different toxic algal species under various conditions.

algicidal microbiomes and provide innovative research avenues to explore individual to community level strategies to control HAB using ecological laws and artificial intelligent approaches.

2. Research methodology

At first, common trends in harmful algal research were identified in response to a variety of biotic and abiotic variables, such as the microbiome. A comprehensive literature search was performed using Scopus, PubMed, Google Scholar and Web of Science databases. The data on the number of studies that used terms such as microbiome, metabolome, nutrient concentration, climate change, land use change, and urbanization was compiled (Fig. 1). The search period covered articles that were published between 2009 and 2023. In this quantitative review article, recently published work was collected and analyzed on algicidal microbial taxa and their properties. The data collected from previous studies have been compiled and presented (see Supplementary Data, Excel sheet). This article classifies these taxa into different microbiome phyla for understanding their role in controlling HABs in aquatic environments. The algicidal activities of different taxa are presented separately within their respective taxonomic groups against the studied algal species. To achieve this, the data was gathered for bacterial phylum *Proteobacteria* from 2000 to 2023, *Actinobacteria* from 2005 to 2023, *Bacteroidota* from 2004 to 2023, and *Firmicutes* from 2002 to 2023 as per published literature (See Supplementary data). This approach was taken to create consistency and logical flow in the data presentation. Additionally, it highlights the collective effectiveness of each microbial group against toxic algae. The mean values were collected and plotted as these values already reported in previous studies. Next, the article recommends emerging biotechnological strategies, such as developing probiotics, engineering synthetic communities, and leveraging machine learning and artificial intelligence approaches, to effectively manage HABs and ensure environmental safety and water quality. The Microsoft Excel was used for initial data organization and analysis, and then R programming (specifically the ggplot2, ggtext package) was used for data visualization. Moreover, Bio-Render was used for creating figures.

3. Results and discussion

Recent research has shown the presence of common bacterial phyla such as Proteobacteria, Planctomycetes, Verrucomicrobia, Bacteroidota, Firmicutes, Acidobacteria, Chloroflexi, and Gemmatimonadetes in freshwater environments. Meanwhile, the rare bacteriosphere includes about 50 phyla; however, a little is known about their functions in freshwater ecosystems (Meirkhanova et al., 2024; Tekle et al., 2021; Zuo et al., 2021). The current review article quantitatively assesses and critically discusses the potential of various microbial species that exhibit significant algicidal activity against different algal species over varying time intervals. It also explores how machine learning approaches can be used to develop synthetic communities for the control and monitoring of HABs in freshwater environments.

3.1. Actinobacteria

Actinobacteria is a dominant phylum of bacteria, and it includes mainly gram-positive bacteria. This phylum is grouped into six different classes that include *Actinobacteria, Coriobacteriia, Acidimicrobiia, Thermoleophilia, Nitriliruptoria* and *Rubrobacteria* (Parte et al., 2012). Only the class *Actinobacteria* is represented by sixteen different orders, among these, the most studied are *Actinomycetales* and *Bifidobacteriales*. Among its order, *Actinomycetales* is confined to the members of the family *Actinomycetaceae* (Barka et al., 2016). These bacteria have the capability to synthesize important biological structures such as mycelium or aerial mycelium, and therefore, these are classically categorized as the filamentous bacteria (Azman et al., 2015). Moreover, these microbes are amongst highly abundant microorganisms in different types of freshwater environments such as rivers, streams, and lakes though their abundance varies depending on climatic and environmental factors such as water nutrient levels, chemistry, and temperature (Hazarika and Thakur, 2020). It is commonly perceived that the abundance of *Actinobacteria* ranges from 50 to 90 percent of total bacterial abundance in some freshwaters under very ideal conditions (Proctor et al., 2018; Song et al., 2019).

The actinobacteria possess highly diverse functional characteristics and thus, they consume simple and highly complex organic compounds that contain essential and limited nutrients such as P, C, and N. Most of these organic substances are synthesized and released by the phytoplanktonic organisms including algae (Eckert et al., 2013). Apart from consuming organic compounds, these microbes require oxygen for their survival (Barka et al., 2016) and reproduction (Taipale et al., 2009). Meanwhile, HABs not only reduce oxygen concentrations but also produce huge quantities of toxic compounds (Hallegraeff, 2003). Therefore to acquire oxygen and reduce organic loads, these bacteria breakdown and inhibit different types of algae including harmful ones in freshwater environments (Mallevialle and Suffet, 1987), and thus, some studies showed a strong relationship between HABs and Actinobacterial abundance in the freshwaters (Berry et al., 2017). The quantitative and critical analysis of previous studies demonstrated a diverse range of highly efficient Actinobacterial species, which were capable of effectively inhibiting the growth of various HAB-forming algae over different time intervals (Fig. 2A). The data showed that the inhibition of various harmful algal species varies from 56 to 100 percent within as less as 6 to 240 h depending on type of algal and algicidal Actinobacterial species. The findings suggest that Actinobacterial species such as *Brevibacterium* were relatively less effective against toxic algal species like *M. aeruginosa*, whereas other species, mostly belonging to *Streptomyces*, demonstrated stronger algicidal activities against a diverse range of algal species, including *M. aeruginosa, Heterosigma akashiwo, Phormidium angustissimum, Alexandrium tamarense, Cochlodinium polykrikoides*, and *Phaeocystis globosa*. This could be because *Streptomyces* are also known to produce algicidal substances that inhibit algal growth in a very short period of time (Fig. 2A, Supplementary Table S1). However, the effectiveness of different *Streptomyces* strains varied across different time intervals, which nevertheless suggest the further screening of efficient algicidal strains from this species. Meanwhile, some studies have also reported the ecological phenomenon of coexistence among Actinobacterial species under oligotrophic conditions (similar to those seen under HABs) (Ma et al., 2022; Yan et al., 2021). Therefore, these bacteria offer a great opportunity to develop probiotics containing either highly efficient individual or mixture of diverse coexisting species to control HABs in the freshwater environments. In conclusion, among *Actinobacteria, Streptomyces* species represent as promising candidates for developing synthetic communities through machine learning approaches (see following sections) to combat HAB invasions in freshwater environments.

3.2. Bacteroidota

The Bacteroidota phylum also represents a huge diversity of bacteria. The members of this phylum have been reported in several ecosystems such as water, soil, plant and animal while these microbes also exhibit diverse mutualistic, pathogenic, and neutral interactions with other higher trophic level organisms (Woese et al., 1990; Wolińska et al., 2017). Bacteroidota group is represented by various classes such as *Bacteroidia, Cytophagia, Flavobacteria*, and *Sphingobacteria*, each having several hundred bacterial species. These types of microbes are gram-negative, and thus they represent diverse physiological diversity in terms of their life-history strategies such as being anaerobic (Bacteroidota) to strictly aerobic (Flavobacteria) in nature (Garrity et al., 2004). These microbes substantially contribute to the abundance of aquatic bacterioplankton, while they play multiple roles in the aquatic environments such as breakdown of contaminants and suppressing pathogenic microbes (Alonso-Sáez and Gasol, 2007; Kragelund et al., 2008). Previous research has reported substantial changes in the composition of bacterial communities, which was largely dominated by the phylum Bacteroidota, under the influence of toxic cyanobacterial blooms (Woodhouse et al., 2016). Scherer et al. investigated temporal dynamics in bacterial communities in response to the toxic cyanobacteria and HABs in aquatic environments. They reported a higher abundance of Bacteroidota in May and June 2015, respectively; thus, predicting the impact of seasonal variations on these microbes. Moreover, the bacterial OTUs belonging to the family *Saprospiraceae* and *Sphingobacteriales* showed a positive correlation with the occurrence of toxic algal species such as *Microcystis,* and these taxa were described as the algicidal bacteria (Scherer et al., 2017). The microbial taxa related to *Saprospiraceae*, are previously described to metabolize various toxins (secondary metabolites) produced by the harmful algal species (McIlroy and Nielsen, 2014). Moreover, the bacteria belonging to the *Sphingobacteriales* are known to lyse various algal species by producing different algicidal metabolites, whereas the members of *Saprospiraceae* could potentially control harmful algae by feeding on them (Lewin, 1997). Previous research showed that *Chryseobacterium* sp. GLY-1106 and *Aquimarina* sp. antisso-27 bacterial strain produce diketopiperazines and l-amino acid oxidase which significantly inhibited the growth of a toxic algae (*M. aeruginosa*) (Chen et al., 2011; Guo et al., 2015). Notably, several Bacteroidota species such as *Arenibacter* sp., *Tenacibaculum* sp., Aquimarina sp. that can inhibit the proliferation of harmful algae from 80 to 100 % within 2 to 168 h, depending on type of algal and algicidal Bacteroidota species and experimental conditions (Fig. 2B). The critical analysis of Bacteroidota species showed that *Aquimarina sp.*, *Pedobacter* *sp*., and *Chryseobacterium sp*. were effective in antagonizing *M. aeruginosa*, while other species such as *Tenacibaculum sp*., *Flavobacterium sp*., and *Maribacter dokdonensis* exhibited antagonistic activity against algal species like *Karenia mikimotoi*. Meanwhile, some bacterial species showed more specific algicidal effects; for example, *Arenibacter sp*., targeted *Akashiwo sanguinea*, whereas *Muricauda sp*., was antagonistic to both *Skeletonema costatum* and *Prorocentrum micans*. Likewise, *Roseivirga sp*., also demonstrated activity against *S. costatum*. Given the high abundance of Bacteroidota taxa in freshwater environments and their demonstrated algicidal potential, this analysis suggests that these bacteria could serve as a valuable natural resource for developing sustainable machine-learning based ecological solutions to address HABs in freshwater systems.

3.3. Cyanobacteria

Cyanobacteria, also known as blue, red, or green algae, are a highly diverse group of aquatic microbes. These microbes are found naturally in freshwater and marine environments (e.g., lakes, rivers, ponds, and oceans). They play a significant role in these environments by regulating primary production and nutrient cycling and also serve as a source of food for higher trophic organisms (Berezina et al., 2021; Sunagawa et al., 2015). Most cyanobacteria can perform photosynthesis using phycobilin or chlorophyll, which is similar to other green plants (Sánchez-Baracaldo and Cardona, 2020). Meanwhile, some cyanobacteria possess highly specialized cells called heterocyst in which they perform nitrogen fixation (Kumar et al., 2010). The cyanobacteria with and without heterocyst or photosynthetic pigments also play essential roles in the health and functioning of freshwater environments (Catherine et al., 2013; Scott and Marcarelli, 2012; Stanier and Cohen-Bazire, 1977). Both biotic and abiotic factors differentially regulate cyanobacterial growth, succession, and establishment in freshwaters depending on site, habitat, and climate-specific conditions (Vanderley et al., 2021; Zhang et al., 2012). Abiotic factors, notably, nutrient concentrations or loading of N and P, could significantly accelerate cyanobacterial growth (Burford et al., 2023). Moreover, light intensity is also a significant factor that regulates the availability of photosynthetically active radiation for cyanobacteria to perform photosynthesis. The availability of light also influences water temperature which can impact the metabolic rates of cyanobacteria (Golden, 1995). That is why climate warming represents a significant boost to the growth of algae, for instance, the growth of harmful algae, in the water bodies (Foysal et al., 2024; Visser et al., 2016). Similarly, the pH of the water can also impact cyanobacterial cellular physiological functions and the availability of trace elements, which are required for different cellular processes (Hinners et al., 2015). Likewise, biotic factors also play a significant role in regulating the growth and proliferation of cyanobacterial growth. Among these, competition for resources (nutrients, space etc.) and antagonistic interactions with other microorganisms can limit the expansion of cyanobacteria. However, synergetic interactions with bacteria, viruses, and protists (microbial loop) may positively impact cyanobacterial growth and proliferation through various symbiotic benefits (Sharma et al., 2011). There are several species of algae or cyanobacteria that cause HABs in the freshwater ecosystems such as *Microcyst* sp., *Pseudochattonella verruculosa* and *Alexandriun catenella* etc., (Gajardo et al., 2023). A previous study investigated whether and how the M. *aeruginosa* (HAB species) can be controlled by the native non-toxic green algal species. Researchers reported that both monocultures and mixtures of various green algal species (*P. tetras, O. polymorpha, M. minutum, P. duplex*, and *S. acuminatus*) substantially suppressed the growth of *Microcyst sp*., thus suggesting that native green algae could potentially suppress HABs in the freshwater environments (Nolan and Cardinale, 2019). This research suggests that promoting the growth of native green algae can help us control or reduce the episodes of HABs in the natural water environments.

3.4. Firmicutes

Firmicutes are ecologically important phylum of gram-positive bacteria that are ubiquities in the water and sediment environments (Reichenberger et al., 2015) whereas bacterial species belonging to this group could be aerobic and anaerobic in nature. These bacteria have been divided into seven classes (Bacilli, Clostridia, Erysipelotrichia, Limnochordia, Negativicutes, Thermolithobacteria, and Tissierellia), 13 orders, 45 families, and 421 genera (Seong et al., 2018). The abundance of *Firmicutes* ranges from 10 to 90 % in freshwaters depending upon on different geographic, climatic, and environmental conditions (Jeon et al., 2017; Wunderlin et al., 2014). Bacteria from this phylum significantly regulate the breakdown of organic materials, thus contributing to nutrient turnover and cycling (Jiang et al., 2019). It is commonly perceived that these microbes play a critical role in the cycling of essential limiting nutrients such as N and P, which are important for other microbes and higher trophic level organisms in aquatic systems (Zhao et al., 2017). Among firmicutes, *Bacillus* are the most prevalent and well-studied group of organisms, and majority of *Bacillus* species *(*>62 %*)* are capable of effectively reducing the growth of bloom-causing cyanobacterial species such as *M. aeruginosa* (Bi et al., 2019). Intriguingly, these bacteria produce specific algicidal compounds to target certain algal species. For instance, *Brevibacillus* sp., produces S-5A (hexahydropyrrolo [1,2-a] pyrazine-1,4-dione) and S-5B (3-isopropyl-hexahydropyrrolo[1,2-a]pyrazine-1,4-dione) to inhibit the growth of toxic *M. aeruginosa* (Li et al., 2015). *Similarly, Paenibacillus polymyxa* produces depsipeptide antibiotic to suppress the growth of *M. aeruginosa* (Kodama et al., 2006). Meanwhile, several *Bacillus* sp., are known to produce a diverse set of algicidal compounds against several types of algal species. Overall, bacteria from this phylum are the most promising microbes because of their algicidal substances to control the blooms of cyanobacteria in natural waters (Supplementary Table S2).

Within the phylum *Firmicutes*, various bacterial species show different levels of algicidal activities with different time points, for instance, *Bacillus subtilis S3* demonstrates 70 percent algicidal activity against *Alexandrium pacificum* in 12 h. On the other hand, *Brevibacillus laterosporus* requires 24 h to exhibit 100 percent algicidal activity against *H. akashiwo* (Fig. 2C). The critical analysis of data on *Firmicutes* suggests that different species of bacillus, such as *B. licheniformis, Paenibacillus sp., B. mycoides, P. alvei, P. polymyxa, B. altitudinis* demonstrated strong algicidal activity against prominent toxic algal species, including *M. aeruginosa*. Additionally, several bacillus strains have shown antagonistic effects against various other algal species such as *A. pacificum, A. minutum, G. catenatum, P. globosa, H. akashiwo,* and *C. marina*. In simple terms, *Bacillus*-derived algicidal bacteria demonstrate remarkable potential as biocontrol agents for developing synthetic microbial communities aimed at mitigating cyanobacterial blooms. The microbial synthetic communities, constructed with AI-driven approaches, can enhance the protection and quality of freshwater systems, as discussed in the following sections.

3.5. Proteobacteria

Proteobacterial phylum represents the gram-negative taxa. This group of microbes covers the most studied bacteria because of their ubiquitous abundance, presence, and functional significance in various ecosystems such as freshwater. It is important to mention that proteobacteria are ecologically classified as, mainly "copiotrophic (with some exceptions)" because they break down and utilize labile organic carbon resources while they perform vital roles in important biogeochemical processes such as nitrogen (fixation, uptake, immobilization, nitrification, denitrification, and mineralization etc.) and cycling of other nutrients such as C, S, P and minerals (Delmont et al., 2018; Eilers et al., 2010; Russo et al., 2016). Owing to their diverse physiological and metabolic characteristics, these microbes are known to regulate the functioning and stability of freshwaters environments (Wang et al.,

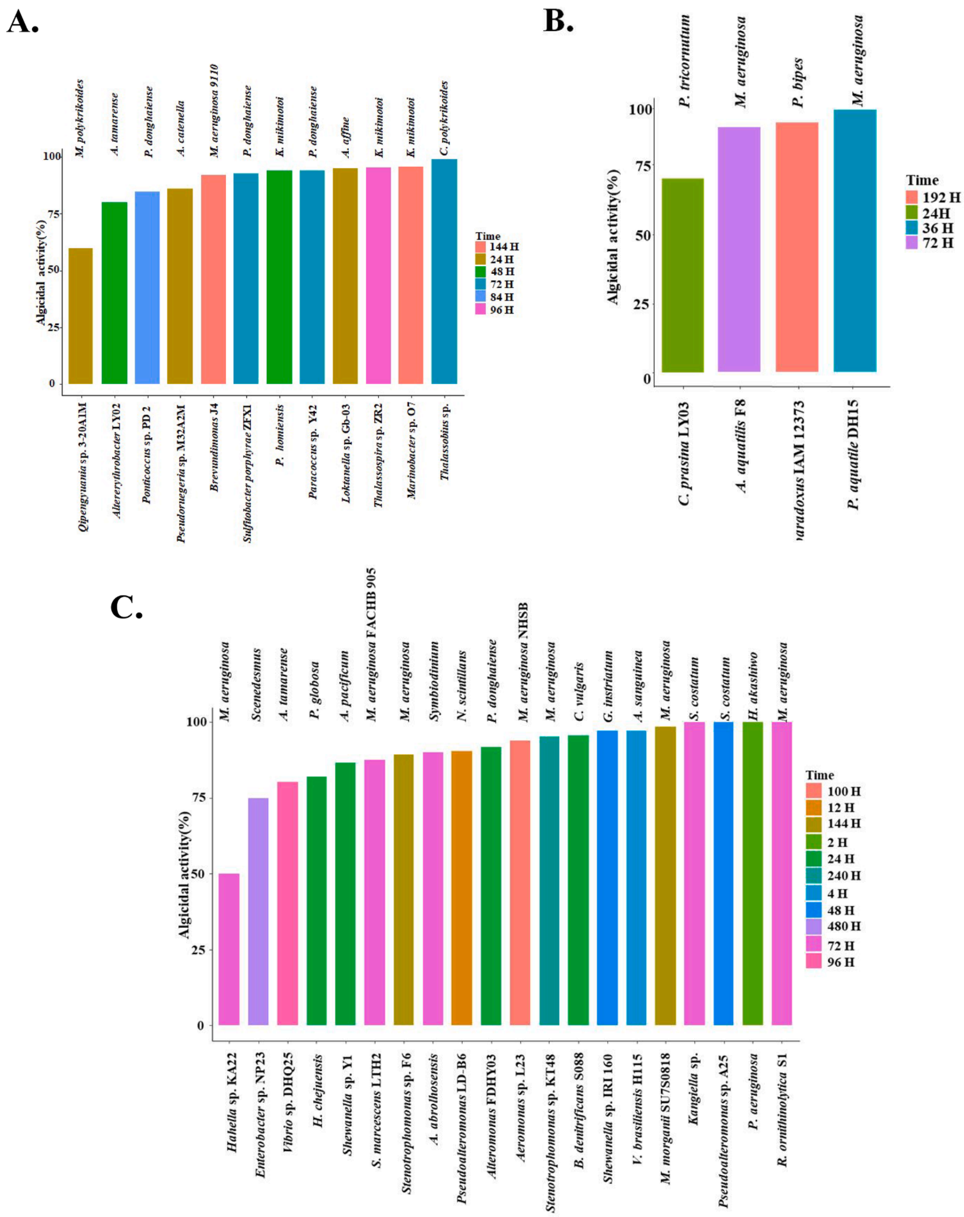

Fig. 3. Bacteria from proteobacteria (a. alpha-proteobacteria; b. beta-proteobacteria and c. gamma-proteobacteria) and their algicidal activity against HAB causing algae.

2020). The *Proteobacteria* generally demonstrate higher dominances in the aquatic ecosystems whereas their abundance varies from 20 to 80 percent of total bacteria in the freshwater environments (Zuo et al., 2021). These bacteria are represented by several taxonomic classes such as alpha- *Proteobacteria*, beta- *Proteobacteria*, gamma- *Proteobacteria*, delta- *Proteobacteria*, epsilon- *Proteobacteria*, and zeta- *Proteobacteria* (Fukuyama et al., 2020). Among these, both beta- *Proteobacteria* and alpha- *Proteobacteria* are known to show a higher abundance in the freshwater environments (Glöckner et al., 2000). The beta- *Proteobacteria* participate in the decomposition of complex organic materials in freshwater ecosystems (Sun et al., 2017). Meanwhile, the gamma- *Proteobacteria* demonstrate a higher abundance in the freshwater environments near residential areas such as lakes, rivers, and streams where nutrient loading is higher (Cheng et al., 2016). These microbes are important in reducing nutrient loads under nutrient-rich or eutrophic conditions caused by HABs (Wang et al., 2020). Recent evidence suggested that some members of alpha- *Proteobacteria* such as phenylobacterium showed a strong positive correlation with the toxic profiles and abundance of *M. aeruginosa* (Zuo et al., 2021). Some researchers have revealed the algicidal activities of several proteobacterial species (*Hahella Sp*., *Pseudomonas, Sp*.,) against HAB forming algal species, by releasing various secondary enzymes and anti-microbial chemicals such as prodigiosin, serine protease and benzoic acids (Supplementary Table S3).

Different groups of *Proteobacteria* show varying levels of algicidal activities against different algal species as discussed below.

3.5.1. Alpha-proteobacteria

These are highly diverse groups of bacteria that play significant roles in reducing HABs by different algicidal activities. First, some alpha-proteobacterial species could exhibit algicidal activities by directly attaching to the algal cells and causing lysis and algal death by producing degenerating enzymes or other compounds (Coyne et al., 2022). Second, these bacteria can also release secondary metabolites that are toxic to HAB causing algae. These algicidal compounds may include proteases, lipases, and other enzymes, which could degrade cellular components through specific toxins that disrupt algal cellular and physiological processes (Imai et al., 2006). For instance, some alpha-proteobacterial species like *Phaeobacter inhibens* can produce reactive oxygen species (ROS) that can damage *Emiliania huxleyi* by causing sever oxidative stresses (Bramucci and Case, 2019). Third, in some cases, alpha-*Proteobacteria* may competently exclude harmful algal species for essential limiting nutrients, such as N, P and S (Kodama et al., 2006). By consuming these nutrients, these microbes could negatively impact the growth, succession and establishment of algal species in the aquatic systems. Four, certain members of alpha-*Proteobacteria* are also capable to induce a programmed cell death (apoptosis) in the harmful algal species. Certain bacteria belonging to alpha- *Proteobacteria* release signaling molecules that can trigger apoptosis-like pathways in the algal cells, thus leading to the process of apoptosis (Bramucci and Case, 2019). Fifth, as a dominant group of freshwater bacteria, these microbes can also regulate the composition, diversity and functioning of aquatic microbial communities. Thus, by reshaping the microbial community structure, they can potentially create highly unfavorable conditions for algal growth and establishment. Current analysis shows that they can suppress the growth of different algal species from 69 to 99 % within 24 to 144 h depending on the type of interacting species and growth conditions (Fig. 3A). The *alpha- Proteobacteria* exhibited remarkable diversity, thus representing species such as *Qipengyuania* sp., *Altererythrobacter, Pseudoruegeria, Brevundimonas, Paracoccus* sp., *Loktanella* sp., and *Thalassobium* sp. all of which showed antagonistic activity against a range of algal species, such as *Margalefidinium polykrikoides, A. tamarense, A. catenella, M. aeruginosa, A. affine*, and *C. polykrikoides*. In addition, species like *Ponticoccus* sp., *Sulfitobacter porphyrae* and *Paracoccus* sp. were found to be antagonistic to *Prorocentrum donghaiense*, while algae like *K. mikimotoi* was effectively controlled by *Paracoccus homiensis, Thalassospira* sp., and *Marinobacter*. This analysis suggests that although alpha- *Proteobacteria* encompass a broad range of algicidal species, only a few have demonstrated activity against highly toxic algae like *M. aeruginosa*. Their overall algicidal diversity, however, highlights their potential to manage mixed-species algal blooms. This diversity, when integrated with machine learning approaches, could facilitate the design of synthetic microbial communities that may address complex, multi-species algal bloom scenarios.

Therefore, considering the significant algicidal potential of alpha-*Proteobacteria*, these microbes represent a great biological resource to develop microbial based approaches for the control of HABs to protect precious water resources.

3.5.2. Beta-Proteobacteria

They demonstrate diverse functional activities in freshwater and thus could suppress algal growth in different ways. For instance, first, beta-*Proteobacteria* are known to release several secondary metabolites like enzymes and bio-active compounds that show algicidal effects in the aquatic environments (Kang et al., 2008). These compounds may inhibit algal growth directly (antibiotics) and indirectly by disrupting algal cellular functions (lytic enzymes) (Li et al., 2016). Second, these bacteria demonstrate a significant level of competition for essential resources such as nutrients and space, and thus they can limit the available resources for algal growth. Thus, the ecological competitive exclusion by these bacteria can indirectly suppress algal colonization, growth, and establishment (Zheng et al., 2024). Third, the beta- *Proteobacteria* can also break down and consume toxins produced by different harmful algal species (Lemes et al., 2008) that may help in reducing the negative effects of HABs on microbial communities, which in return, could serve as a biological buffer against algal invasions. Recent laboratory research has shown that some species of beta- *Proteobacteria* possess algicidal properties and are capable of reducing the growth of various algal species by 70 to 95 percent within 24 to 192 h depending on experimental conditions (Fig. 3B). This analysis revealed that relatively few beta- *Proteobacteria* species, such as *Alcaligenes aquatilis* and *Paucibacter aquatile*, exhibited algicidal activity against *M. aeruginosa*. In contrast, *Chitinimonas prasina* and *Variovorax paradoxus* demonstrated effective algicidal activity against *Phaeodactylum tricornutum* and *Peridinium bipes*. Although reports of algicidal activity within this bacterial group are limited, these findings highlight promising opportunities to further explore their potential for controlling HABs in freshwater ecosystems.

3.5.3. Gamma-proteobacteria

These bacteria are amongst the largest group of aquatic microorganisms, which play crucial roles in reducing HABs in freshwater by different means. First, some species of gamma- *Proteobacteria* synthesize catalytic enzymes that can break down algal cell walls through lysis (Lin et al., 2016; Wang et al., 2005). Second, these bacteria can capture and sequester essential nutrients (N, P, S), light, and space (Mayali and Doucette, 2002). By competently excluding algae out of ecological niche and essential resources, they can suppress various algal species (Liu et al., 2014). Third, gamma- *Proteobacteria* also release secondary metabolites which can toxify and inhibit algae, thus directly suppressing the colonization and establishment of HABs in freshwaters (Wang et al., 2005). Fourth, these bacterial subphyla also develop aggregates and flocs that can suppress algae across water columns. By doing this, these bacteria can limit algal access to several resources such as light and nutrients (Gallardo-Rodríguez et al., 2019). Moreover, the meta-analysis findings reveal that several species of gamma- *Proteobacteria* are known to limit algal growth by their algicidal effects on several toxic species including *M. aeruginosa* and others, from as low as 50 to as high as 100 percent within 4 to 480 h, depending on experimental conditions (Fig. 3C). Several gamma-Proteobacterial species, including *Hahella sp.*, *Serratia marcescens, Stenotrophomonas sp., Aeromonas sp., Morganella morganii*, and *Raoultella ornithinolytica*, exhibited a wide range of algicidal activities against *M. aeruginosa*, from low to high efficacy. Other species, such as *Kangiella sp.* and *Pseudoalteromonas sp.* also demonstrated notable antagonistic potential against *S. costatum*. In addition, certain Gamma-proteobacteria—including *Enterobacter sp., Vibrio* sp., *Hahella chejuensis, Shewanella* sp., *Alteromonas abrolhosensis, Pseudoalteromonas sp., Bowmanella denitrificans, Vibrio brasiliensis*, and *Pseudomonas aeruginosa*—showed species-specific algicidal activity against a variety of harmful algae, such as *Scenedesmus, A. tamarense, P. globosa, A. pacificum, Symbiodinium, Noctiluca scintillans, Prorocentrum donghaiense, Chlorella vulgaris, Gonyaulax instriatum, Akashiwo sanguinea* and *H. akashiwo*. These findings highlight the remarkable algicidal diversity within gamma- *Proteobacteria*, making them strong candidates for the development of synthetic microbial communities. When combined with machine learning approaches, these bacteria offer significant potential for targeted control of harmful algal blooms.

3.6. Verrucomicrobia

The *Verrucomicrobia* are unique group of uncultured bacteria that were lately described with the advent of 16S rRNA gene sequence (Hedlund et al., 1997). Then, these microbes were grouped into a separate phylum (Pitt et al., 2020). These are common free-living microorganisms, and have been reported in several habitats ranging from water (freshwater, marine) to soil, and animal gut environments (Wagner and Horn, 2006). Although these microbes are represented by large number of unclassified taxa, currently, about 50 species and 20 genera are grouped under this phylum (Pitt et al., 2020). Moreover, these taxa are abundant in all freshwater ecosystems though their abundance vary with water properties. For instance, water ecosystems with humid conditions show a high abundance of these taxa than less humid waters (Arnds et al., 2010). In addition, these microbes are classified as oligotrophic microbes (nutrient poor conditions), thus suggesting that they can break down complex and recalcitrant organic substances due to their diverse metabolic activities. Given the ubiquitous abundance of *Verrucomicrobia* and their active involvement in several biogeochemical processes, they could be important during HABs in the aquatic ecosystems. These microbes have been associated with

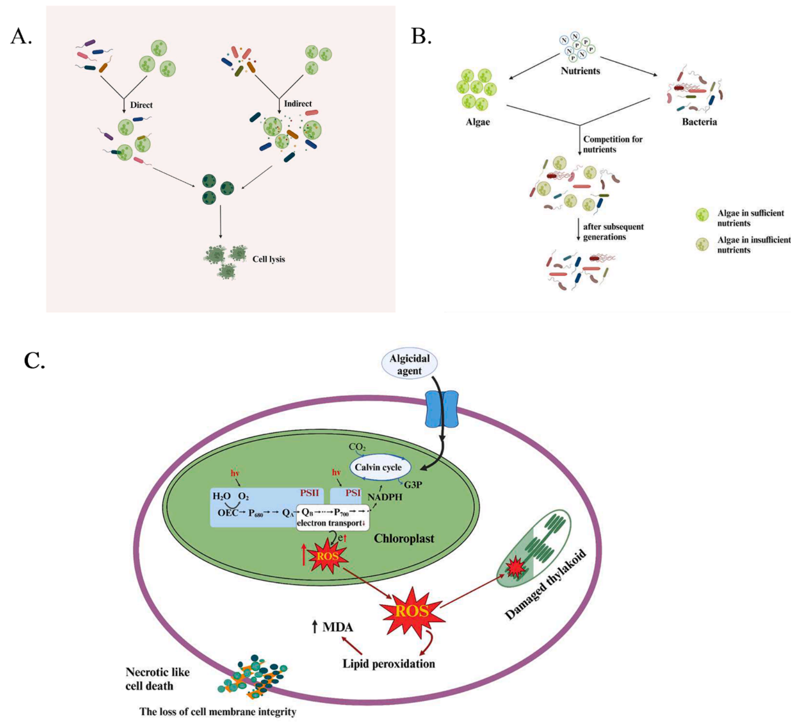

Fig. 4. Different mechanisms by which bacteria suppress algal growth. a. The lysis and mortality of algal cells by algicidal bacteria. Modified from (Yang et al., 2020a). b. Ecological competition for essential nutrients by native bacteria suppresses algal growth in the water environment. c. The mechanism of algal cell death under elevated ROS caused by native bacterial taxa. Modified from (Wu et al., 2022; Yang et al., 2017).

algal blooms because these algae produce highly complex organic compounds (Newton et al., 2011). Therefore, some studies have reported that the abundance of *Verrucomicrobia* increases with an increasing degradation of *Microcystis* in the aquatic ecosystems (Shao et al., 2013). There is emerging evidence that *Verrucomicrobia* can contribute to controlling HABs in freshwater environments by different ways. First, *Verrucomicrobia* is involved in the degradation of complex organic materials such as polysaccharides. This process can contribute to the cycling of limiting nutrients (N, P) in aquatic environments, thus favoring the growth of bacteria against algal colonization, growth and blooms in freshwaters. Second, *Verrucomicrobia* can potentially exclude algae out of ecological niche by utilizing essential nutrients and minerals. By effectively consuming available nutrients, *Verrucomicrobia* could limit the supply and availability of nutrients to the algae, thereby decreasing the growth and blooms of harmful algal species (Cardman et al., 2014; Orellana et al., 2022). Third, some species of *Verrucomicrobia* may demonstrate antagonistic effects on harmful algal species by producing biochemicals that may inhibit algal growth (Di Costanzo et al., 2023). Fourth, *Verrucomicrobia* is an integral component of aquatic bacterial communities and thus, they can maintain ecological balance and interactions within microbial communities, which in return, can serve as a buffer against harmful algal invasions. Nevertheless, the role of *Verrucomicrobia* in controlling HABs is still an understudied area of research. Therefore, further research could pursue their roles in controlling algal growth by nutrient deprivation, competitive exclusion, and antagonistic interactions to manage freshwater environments.

3.7. Miscellaneous bacterial phyla

There are several other bacterial phyla (e.g., *Acidobacteria, Planctomycetes*) that exist in various freshwater environments though their roles in controlling HABs are not well studied. Among these, *Acidobacteria* are diverse group of mostly uncultured bacteria, and they have been reported in different habitat types including freshwater environments. These bacteria are known to survive under nutrient-limited conditions, which nevertheless, predict their capability to competently exclude algae out of ecological niche. Similarly, *Planctomycetota* are also a widely distributed group of gram-negative bacteria that have the ability to perform several biological processes such as anammox, nitrogen fixation and decomposing algal polysaccharides, and are often seen in association with HABs in different environments (Cai et al., 2013; Sidhu et al., 2023; Suarez et al., 2023). For instance, their role in anaerobic ammonium oxidation (anammox) can help in suppressing harmful algae. They convert labile nutrients such as ammonium and nitrite into nitrogen gas (Teixeira et al., 2014). By doing so, these bacteria can limit the availability of nitrogen to algal species that can contribute to reducing HABs in freshwater.

4. Future implications

The findings can be helpful in developing microbial based biotechnologies to control the HABs in freshwater.

4.1. Using individual strains as environmental probiotics

Future research could shortlist highly efficient strains to develop environmentally compatible microbial probiotics to treat freshwater environments against HABs invasion. Using individual algicidal bacterial species to control HABs in freshwater could prove a highly practicable biocontrol strategy. This critical analysis suggests that, depending upon environmental conditions, one can select beneficial bacterial species with desired algicidal mechanisms, such as those (i) directly lysing algal cells (enzyme release)(Chen et al., 2011; Shi et al., 2006), (ii) competing for nutrients and depriving algae (Jianhong and Shaobin, 2002), (iii) micropredation (feeding algae) (Song et al., 2020), (iv) forming biofilms (Sun et al., 2023), (v) improving microbiome community and functioning (Gajardo et al., 2023), (vi) oxidative stress induction (Wu et al., 2022)and (vii) algicidal compounds (focused on this review) (Fig. 4ABC).

To implement this strategy, researchers and environmental

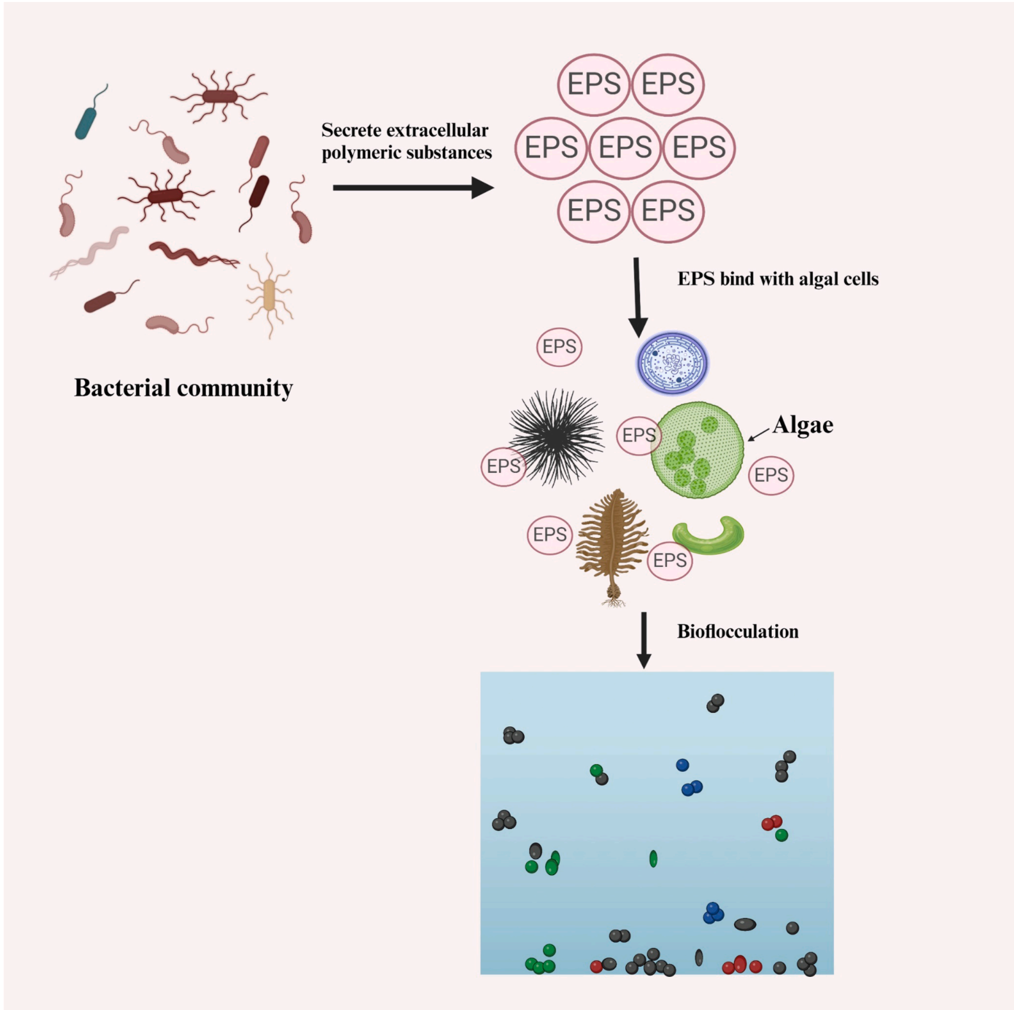

Fig. 5. Bio-flocculation of algae by bacterial EPS inhibits the growth of toxic algal species. Modified from (Anabtawi et al., 2024).

biotechnologists must select highly efficient native bacterial strains for field trials and application while considering their survival as well as colonization potential and biosafety concerns for non-target species and other environmental resources.

4.2. Using mixed strains to develop synthetic communities

Meta-analysis also provides a strong basis to develop mixed culture synthetic communities that can be used to treat the HAB contaminated freshwater environments. Synthetic microbial consortia have previously been used in various biotechnological processes and applications, including improving plant growth (Wang et al., 2022), nutrient removal (Jia et al., 2016), pathogen control (Jia et al., 2016; Song et al., 2014), bioremediation (Adamu et al., 2023), and host fitness (Song et al., 2014). However, their potential application for controlling HABs in freshwater environments remains largely unexplored. Therefore, it is suggested that mixed synthetic communities could inhibit algal growth by more than one potential algicidal mechanism. The use of synthetic microbial communities may also lead to a bio-flocculation of algal species in freshwater environments. The microbial bio- flocculation is an ecological process by which bacterial communities could suppress HABs by attacking the algal cells in the aquatic environments (Fig. 5) (Alam et al., 2016).

However, developing mixed culture synthetic bacterial communities to control HABs in freshwater involves several systematic steps. First, it is important to identify the algal species that are causing HABs and understand physicochemical properties (pH, chemistry, temperature, nutrient concentrations) of the target water body. Second, one would select the most efficient culturable bacterial taxa from the contaminated waterbody that possesses either of the abovementioned algicidal mechanisms to inhibit algal growth. Third, test the growth of bacterial strains in different mixed combinations and attempt to find combinations that show more synchrony or niche partitioning among bacterial species to ensure their viability, colonization, coexistence, and effectiveness. After identifying the optimal mixed culture combinations, it is also important to determine the appropriate concentrations, the most effective application methods, and any potential environmental risks associated with introducing these synthetic communities into natural systems. The review article proposes that introducing synthetic microbial communities in the early stages of HAB development could yield promising results, much like early-stage treatment of infections in patients. Nevertheless, selecting the method of introducing synthetic microbial communities into contaminated water is of utmost importance depending on environmental conditions. For instance, direct one-time inoculation or slow-release formulations can be chosen to ensure sustainable colonization, succession and establishment of anti-algal microbial communities in the freshwater environments.

4.3. Using artificial intelligence to monitor bacterial effects on HABs

The robust understanding of HABs and algicidal bacteria can help in accurate forecasting and mitigation of toxic algae in the freshwaters.

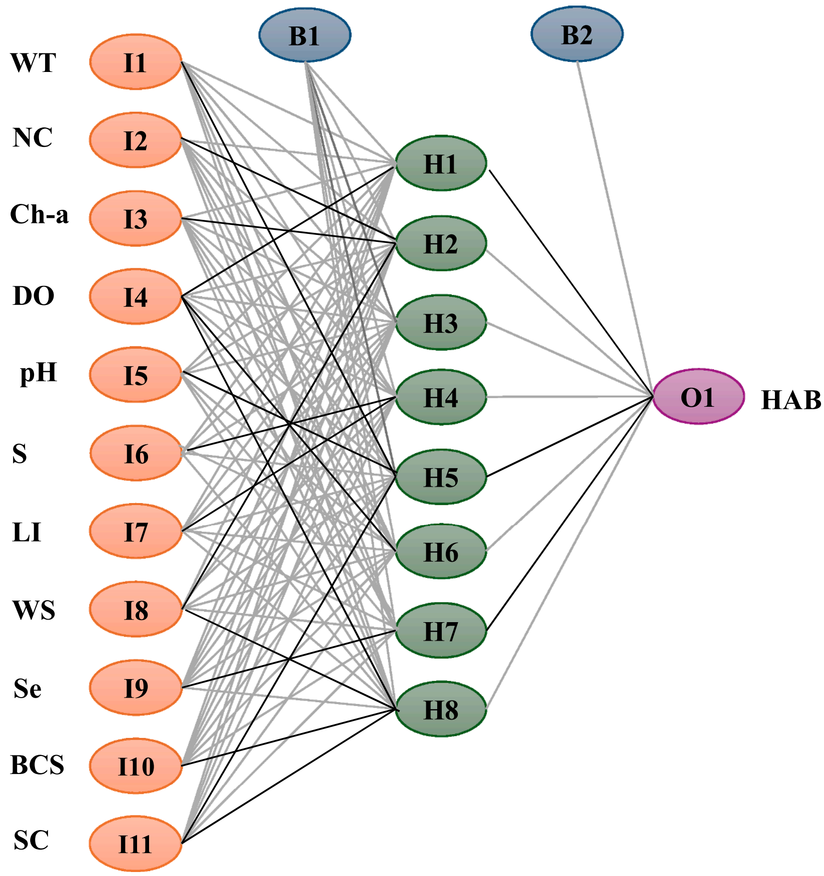

Fig. 6. A theoretical framework of using ANNs in monitoring HABs and algicidal taxa for water quality. Artificial neural interpretation diagram with 11, 8, and 1 neuron(s) in the input (I), hidden (H), and output (O) layers, respectively. Bias terms B1 and B2 are introduced to the hidden and output layers. Positive and negative connections are represented by black and grey lines, respectively. Notes: WT = water temperature; NC = nutrient conc.; Ch-*a* = chlorophyll-a; DO = dissolved oxygen; *S* = salinity; LI = light intensity; WS = wind speed; Se = seasonality; BCS = bacterial community structure; SC = species composition; HAB = harmful algal bloom (HAB or no HAB or a probability indicating the likelihood of a bloom).

Thus, the integration of HABs and bacterial data into models has the potential to provide insight into the formation of blooms and the corresponding responses in bacterial communities. In this direction, a wide variety of machine learning algorithms have been used such as random forest (Kehoe et al., 2012; Liu et al., 2023a), artificial neural networks (ANNs) (Abdullah et al., 2022; Baek et al., 2022; Orenstein and Beijbom, 2017), support vector machines (SVMs) (Ribeiro and Torgo, 2008; Vilas et al., 2014) and others (Wen et al., 2022). Random forest (RF) is a machine learning model that makes predictions by using a number of decision trees where each of which is trained on a different subset of data (Biau and Scornet, 2016). In order to enhance the model's accuracy, RF combines the output from every single tree and then uses majority vote for classification and average results for regression to choose the final outcome. RF is faster and offers more precise interpretation for small sets of data as well as noisy, unbalanced, and tabular data (Biau and Scornet, 2016). The ANN is a multifaceted machine learning algorithm that is composed of interconnected nodes that possess a robust capacity to comprehend nonlinear relationships in large-scale data (Kang et al., 2015). The ANNs can process large datasets (algal, bacteria, chemical and climate), and thus can help in interpreting environmental patterns like water temperature and nutrient levels to predict changes in the abundance of toxic algae and alterations in bacterial communities. Using remote sensing and image analysis, ANNs can identify harmful algae species in large waterbodies, thus enabling their accurate and rapid detection. These models can also help in explaining interactions between algicidal bacteria and algae under different environmental conditions from large-scale data set, thus precisely predicting HABs in freshwater environments. More specifically, the ANNs could be valuable tools to optimize synthetic microbial communities, develop highly efficient microbial combinations to control HABs, and develop and refine in situ application strategies. Using a neural network model optimized by machine learning to predict interactions between water properties, microbes, and toxic algae in response to climatic variations, may involve several key steps (Lee and Lee, 2018; Yan et al., 2024). First, data of water properties, microbes, and algae could be collected. Then, the normalization and feature engineering of the collected data will be performed. Afterwards, one can test process that data using ANNs and followed by the model optimization to build future predictions. The large data tools such as PyTorch library in Python can be used to construct the artificial neural networks (Liu et al., 2023b). Finally, it is suggested to use a 4-layer network with ReLU and sigmoid activation functions between hypothetical data layers (input/output), as shown (Fig. 6) to demonstrate the complex interactions between algicidal taxa, toxic algal species and water properties to predict HABs and water quality. Many studies have already applied RF and ANN to predict harmful algal blooms (Table 1). The strengths of both ANN and RF can be used to create a hybrid model that increases prediction accuracy. To reduce input complexity and improve the ANN's learning efficiency, RF can first identify the key features that cause HABs and then utilize those features to train an ANN. By harnessing the power of ANNs to process large, complex datasets, researchers and environmentalists can gain deeper insights into the factors that drive HAB formation, predict and detect bloom more accurately, and develop targeted strategies for controlling them. Moreover, combined techniques may be employed, wherein predictions from both ANN and RF are averaged or integrated via voting mechanisms to produce a final, more resilient and

Table 1
A summary of ANN and RF for predicting HABs in freshwater.

Model name	Input variables	Target variables	Prediction performance	Reference
RF and ANN	1556 observations with 16 attributes from the biological, physical, and chemical categories (1986–2018)	Chl-a	RF (MAE: 0.4453; RMSE: 0.5686; MSE: 0.3233), ANN (MAE: 0.5607; RMSE: 0.6359; MSE: 0.4044)	(Rostam et al., 2021)
ANN	Ammonia, Chlorophyll 1, Chlorophyll 2 and Nitrate	harmful algal blooms	MSE: 0.061; R^2: 0.939	(Yu et al., 2021)
ANN	EC, DO, WT, TN, BOD, COD, TP, TSS, CHL-a and transparency	algal chlorophyll-a and transparency (water clarity)	RMSE: 1.56 to 4.96; R^2: 0.40 to 0.60; MAE: 1.15 to 3.20	(Mamun et al., 2019)
ANN	discharge, temperature, salinity, ΔT, ΔS, transparency, PAR, DS, SR, NO_2^- + NO_3^-, NH_4^+, DSi, PO_4^{3-}, and DIN/DIP (2008–2018)	chlorophyll a	SSE $\leq$ 0.0003; RMSE $\leq$ 0.0173; $R^2 \geq$ 0.9952	(Park and Sin, 2021)
RF	water depth, surface PAR, wind speed, wind direction and benthic light	benthic light affected by L. *majuscula* blooms	R^2 values of the order 0.8	(Kehoe et al., 2012)
RF	physicochemical, hydrological, and meteorological variables	phytoplankton community	R^2: 0.744; RMSE: 0.193; NRMSE: 10.0 % for total biomass	(Liu et al., 2023a)
RF	water temperature, pH, electrical conductivity (EC) and system battery (three years)	Chl-a	MAE: 4.40 and 5.25	(Mozo et al., 2022)
ANN	water quality and meteorological data (1999–2012)	Chl-a	NMSE: 0.003 - 0.005; Corr: 0.53 – 0.895; NSE: 0.633 – 0.808	(Liu et al., 2015; Mozo et al., 2022)
DNN	water temperature, pH, DO, BOD_5, COD, SS, TN, TP, TOC, Chl-a and the number of cyanobacteria	harmful algal bloom	R^2: 0.79; MAE: 0.1; RMSE: 0.2	(Yang et al., 2020b)
RF	Moderate Resolution Imaging Spectrometer (MODIS) data	algal biomass	R^2: 0.85; RMSE: 6.88; MAE: 1.24	(Lai et al., 2023)
RF	TP, TN	Chl-a	R^2: 0.9997; RMSE: 0.0010	(Huang et al., 2022)
RF and ANN	Water quality and meteorological data	Chl-a	RF (R^2: 0.952; RMSE: 3.152; MAE: 1.539) and ANN (R^2: 0.750; RMSE: 6.864; MAE: 3.927)	(Kim and Ahn, 2022)
RF	BOD, COD, SS, E. coil, TN, TP, EC, TOC, W.T., pH, DO, Chl-a, HRT, W. Volume, rainfall, inflow, and outflow (8 days interval)	cyanobacteria	For the same day accuracy is 0.831 and after 7 days, accuracy is 0.825	(Jo et al., 2023)

trustworthy forecast. Since it began, as technology evolves and more data becomes available, the application of RF and ANNs would increase and play central role in the biotechnological monitoring and the effective mitigation of HABs as the function of algicidal microbial taxa, thus safeguarding both freshwater environments.

5. Conclusion

Currently freshwaters are witnessing more frequent HABs due to climatic changes and anthropogenic activities that promote the growth of toxic algae there. However, the growth and development of HABs are mostly determined by ecological interactions among microbial communities, toxic algae, and environmental conditions such as nutrient concentrations, temperature, light intensity and other water physicochemical properties. Meanwhile, there are some native bacterial taxa that act as natural biological buffers against algal invasions and help in limiting the algal growth by different ecological and biochemical mechanisms, including the production of algicidal chemicals. Current critical review article digs the data of algicidal microbial taxa and precisely presents its algicidal potential against toxic algal species. The quantitative analysis shows that algicidal taxa is represented by highly diverse microbial species which are capable of suppressing different types of algal species in a very short period of time. We propose integrated synthetic microbial community and machine learning methodological approaches that can be utilized to address the issue of HABs in freshwater. In summary, the use of native algicidal microbial taxa offers a promising avenue for developing microbial-based biotechnologies to mitigate HABs in freshwater systems. The review article pinpoints several critical research gaps and proposes future research endeavors. First, systematic screening and characterization of native algicidal microbes across diverse freshwater resources are needed to build a robust catalog of functional candidates. Second, the development of synthetic microbial communities tailored for HAB control would benefit from machine learning-guided design strategies that optimize microbial interactions, stability, and efficacy under field conditions. Third, there is a pressing need to integrate real-time environmental monitoring with predictive modeling frameworks, thus using machine learning and remote sensing data to monitor HABs and guide timely biotechnological interventions. Finally, future research should prioritize standardized methodologies for evaluating algicidal performance, scalable delivery systems for microbial consortia, and regulatory frameworks for environmental safety and public acceptance. Addressing these research gaps will be key to translating emerging microbial technologies into practical field-deployable solutions for freshwater ecosystem protection.

CRediT authorship contribution statement

Md Imam ul Khabir: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Nur E. Alam:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **J.F. Jhumur:** Writing – original draft, Software, Methodology, Investigation. **Tanzeel U. Rehman:** Writing – review & editing, Supervision, Software. **Sapna Jain:** Writing – review & editing, Visualization, Validation, Supervision, Resources. **M.A. Rahman:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Methodology. **Yongpo Zhao:** Writing – review & editing, Writing – original draft, Resources, Methodology. **Muhammad Saleem:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

Authors declare no conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.teengi.2025.100030.

Data availability

I have shared all of my data to the attached step.

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